

DAY-3

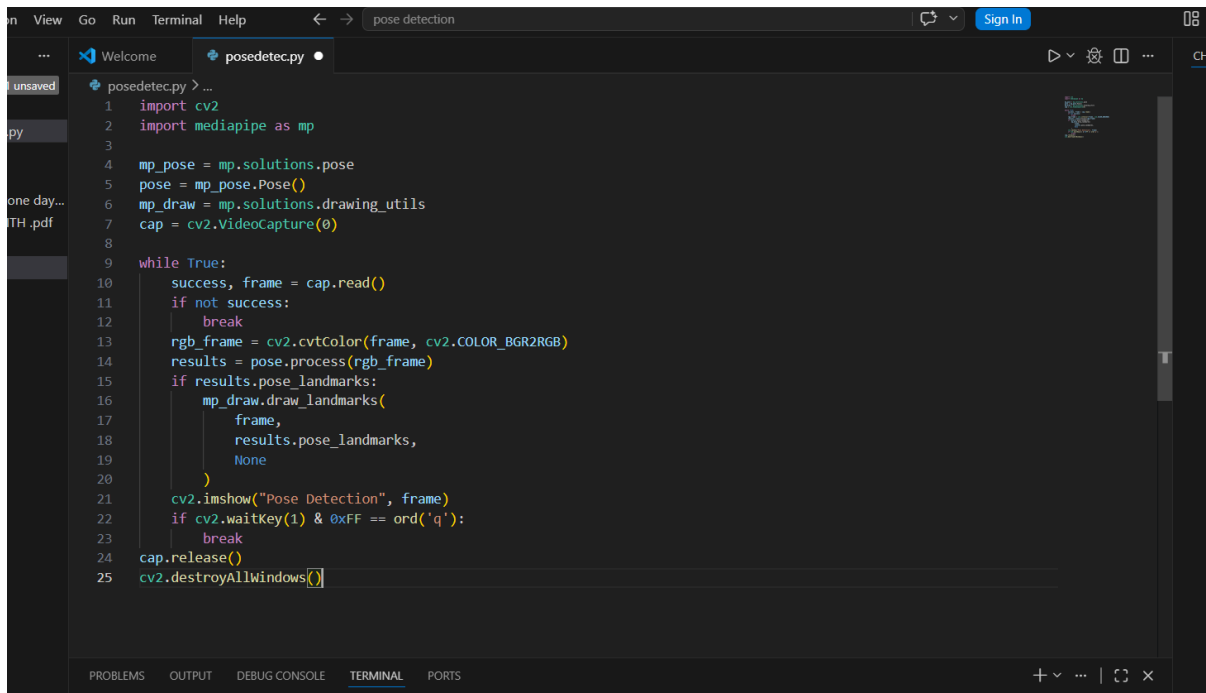
TASK 1 :

HUMAN POSE DETECTION WITH SKELETAL LINES:

On day 2 of this internship, I created a demo to show the working of human pose detection through body landmarks using MediaPipe and OpenCV. Today, I learned to join those body landmarks to create what seems like a skeleton to get a better structure of a human body.

I learned that we just have to change one line from the original human pose detection code yesterday to achieve this.

THE ORIGINAL CODE:



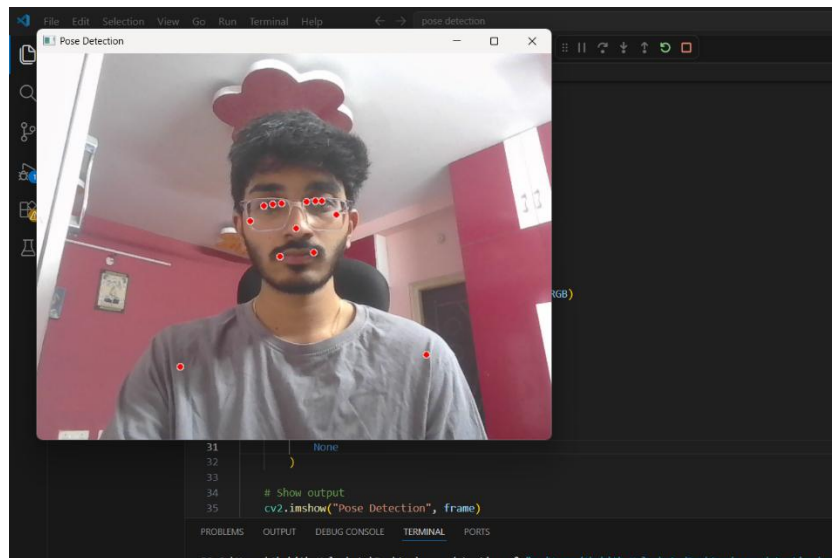
```
1 import cv2
2 import mediapipe as mp
3
4 mp_pose = mp.solutions.pose
5 pose = mp_pose.Pose()
6 mp_draw = mp.solutions.drawing_utils
7 cap = cv2.VideoCapture(0)
8
9 while True:
10     success, frame = cap.read()
11     if not success:
12         break
13     rgb_frame = cv2.cvtColor(frame, cv2.COLOR_BGR2RGB)
14     results = pose.process(rgb_frame)
15     if results.pose_landmarks:
16         mp_draw.draw_landmarks(
17             frame,
18             results.pose_landmarks,
19             None
20         )
21     cv2.imshow("Pose Detection", frame)
22     if cv2.waitKey(1) & 0xFF == ord('q'):
23         break
24     cap.release()
25     cv2.destroyAllWindows()
```

This is the code that shows the output with body landmarks but no lines connecting the landmarks.

If we change one line of code in the “if results.pose_landmarks:” condition from “none” to “mp_pose.POSE_CONNECTIONS” then it will automatically connect the body landmarks.

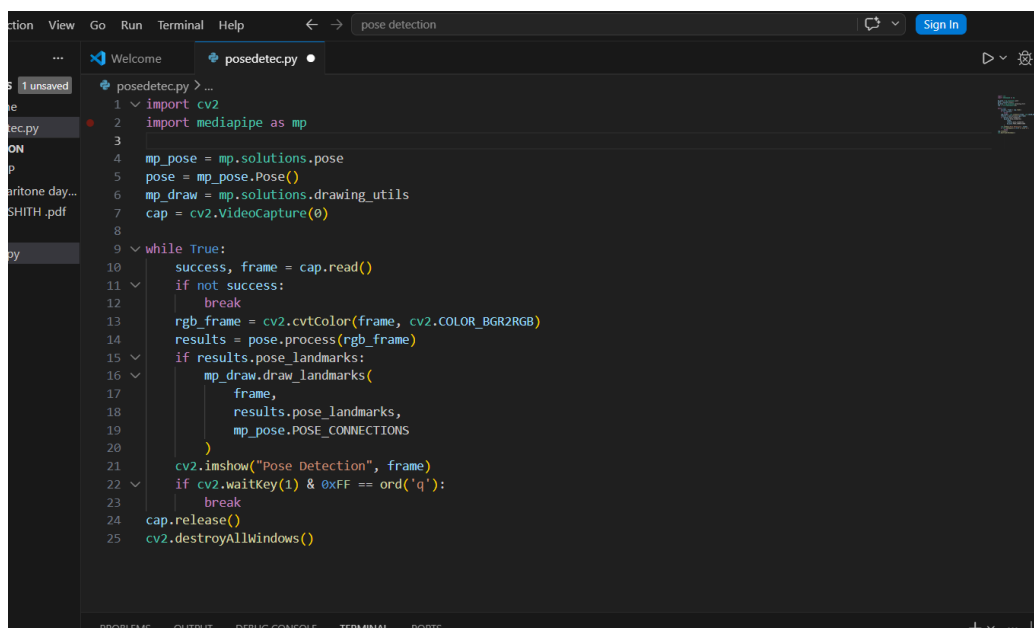
In MediaPipe, “mp_pose.POSE_CONNECTIONS” means A predefined list of landmark pairs that should be connected with lines. All landmarks are assigned a number and then connected according to those numbers.

OUTPUT FOR ORIGINAL CODE: (NO LINES)



As we can see there are no lines joining the landmarks hence it can not be used for body measurements. We require those lines to calculate the distance between those points to get measurements like shoulder width, waist width, head to toe length to estimate height and body shape.

The new edited code to get the skeletal lines between the landmarks by adding `mp_pose.POSE_CONNECTIONS` is given below:



OUTPUT FOR THIS CODE:

